Data Science laboratory

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Subject Code : CGI1242

Subject Course: Data Science

TITTLE: Forecasting Gold Prices using historical Data

OBJECTIVE:

* To build regression Models that forecast the next day's gold Price from historical data and compare them using Mean Squared Error (MSE)

DATASET DESCRIPTION:

* The data set (Gold-Price.csv) has 2,511 daily records from 29 Jan 2016 to 23 Jan 2026, with 16 columns including open, High, Low close, volume, moving averages and volatility-20. The target is the close Price. close and volume have No missing values. Missing values appear only in calculated columns because the first row lack enought Past data.

APPROACH AND TECHNIQUES:

* Dates were sorted and rows with Missing value were removed (2,490 rows left). Lag features were created to avoid data leakage, and a chronological 80/20 Split was used because Shuffling would expose future prices.

* Simple regression and Multiple regression were compared using MSE and a native baseline.

KEY FINDINGS

* Simple regression MSE = 12.86 and Multiple regression MSE = 13.04, so the Simple model performed slightly better. The naive baseline (12.69) performs about the same.

* Gold Price rose from 106.95 (2016) to 458.00 (2026); predictions follow the trend but lag during Sudden changes.

CONCLUSION

* The Simple linear regression Model gave the best forecast (MSE 12.86). It captures the overall trend but cannot handle Sudden or non-linear changes. Time Series Models such as ARIMA or LSTM can be used in future to get better forecasts.